

Notes: Iliev (JF, 2010)

The Effect of SOX Section 404: Costs, Earnings Quality, and Stock Prices

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1 Big picture

- **Research question.** What were the causal effects of SOX Section 404 on small public firms? In particular, did the rule improve reporting quality enough to justify its compliance costs?
- **Core policy setting.** Section 404 required managers to report on internal controls over financial reporting and required an outside auditor to attest to management's assessment. The controversial part for small firms was the auditor attestation and the fixed-cost nature of compliance.
- **Empirical idea.** The SEC implementation rule created sharp public-float cutoffs. U.S. firms above \$75 million in public float had to comply earlier, while firms below the cutoff could delay. Foreign firms faced a separate \$700 million cutoff for auditor attestation.
- **Identification strategy.** The paper compares firms just above and just below the cutoff using a regression discontinuity design. Because some firms may have manipulated public float after the rule became known, the paper instruments 2004 compliance with whether public float exceeded \$75 million in 2002.
- **Main finding on costs.** Section 404 compliance sharply increased audit fees. In the preferred IV specification, filing a management report is associated with roughly a 0.98 log-point increase in audit fees, or about \$698 thousand for the mean firm.
- **Main finding on reporting.** Compliance made reported earnings more conservative: MR filers reported lower accruals and lower discretionary accruals.
- **Main finding on valuation.** The market reacted positively to delays or cost reductions in implementation and negatively to evidence that the SEC would enforce the rule. Over the implementation period, firms predicted to comply underperformed firms predicted not to comply.
- **Economic takeaway.** SOX Section 404 appears to have improved earnings quality, but for small firms the direct and indirect costs were large enough that the net market-value effect was negative.

2 Incremental contribution of the paper

- **From broad SOX debate to a clean quasi-experiment.** Earlier SOX studies were confounded by Enron-era scandals, the collapse of Arthur Andersen, changing enforcement, and broader scrutiny of corporate reporting. Iliev isolates one part of SOX by exploiting the SEC's staggered implementation rule.
- **Regression discontinuity in regulation.** The paper uses a well-defined compliance threshold rather than comparing all pre-SOX and post-SOX firms.
- **Treatment and control firms are local.** Firms near the \$75 million public-float cutoff are more comparable than the average accelerated and nonaccelerated filer.
- **Explicit evidence on costs, benefits, and net value.** The paper studies audit fees as a cost measure, accruals as a reporting-quality measure, and stock returns as a market-based net-benefit measure.

- **Manipulation-aware design.** The paper documents bunching below the cutoff after firms learned the rule, then uses the predetermined 2002 cutoff as an instrument for later compliance.
- **Second quasi-experiment.** Foreign firms provide a related test because auditor attestation was delayed for foreign private issuers below a \$700 million public-float cutoff.

3 Institutional setting, data, and measurement

3.1 SOX Section 404 and the compliance cutoff

SOX was enacted in 2002 after a series of major corporate scandals. Section 404 required firms to assess internal controls and required auditors to attest to management’s assessment. The SEC implemented the rule gradually. For U.S. firms, the key distinction was accelerated filer status: firms with public float above \$75 million became accelerated filers and had earlier Section 404 compliance obligations.

For fiscal years ending on or after November 15, 2004, accelerated filers had to file a management report and obtain an auditor attestation. Firms below the cutoff could delay. The rule also shortened annual-report deadlines for accelerated filers, which is a potential confounding feature that the paper checks separately.

3.2 Sample construction

The main U.S. sample is built from firms near the \$75 million cutoff. The paper starts with companies whose market capitalization puts them plausibly near the cutoff, hand-collects public float and filing status from 10-K filings, excludes foreign and financial firms, and focuses on firms with 2004 public float between \$50 million and \$100 million. This yields the primary 2004 regression-discontinuity sample.

3.3 Outcome variables

- **Audit fees.** Audit and audit-related fees from Audit Analytics measure the direct monetary cost of compliance.
- **Accruals.** Accruals are earnings minus cash flows. Lower accruals are interpreted as more conservative reporting.
- **Discretionary accruals.** Discretionary accruals are constructed using a cross-sectional Jones model and are intended to isolate the more discretionary component of earnings.
- **Stock returns.** Event-window abnormal returns and longer-run factor-adjusted portfolio returns measure the market’s assessment of the net costs and benefits.
- **Foreign-firm outcomes.** For foreign firms, the paper studies audit fees, accruals, discretionary accruals, and event returns around the auditor-attestation cutoff.

4 Methodology and empirical specifications

4.1 Regression discontinuity around the public-float cutoff

The core cross-sectional specification is:

$$\text{Dep. Var.}_i = \beta_0 + \beta_1 MR_i + \beta_2 PFL_i + \beta_3 PFL_i^2 + \beta_4 PFL_i^3 + \Gamma' X_i + \epsilon_i, \quad (1)$$

where MR_i equals one if firm i files a management report, PFL_i is public float, and X_i contains controls. The coefficient β_1 is interpreted as the local effect of Section 404 compliance for firms near the cutoff.

4.2 Instrumental variables strategy

Because firms could manipulate their public float after learning how the rule would be implemented, the paper instruments actual 2004 MR filing with whether the firm was above the \$75 million public-float cutoff in 2002:

$$MR_i = \alpha_0 + \alpha_1 \mathbf{1}\{PFL_{i,2002} > 75\} + \Theta' X_i + \eta_i, \quad (2)$$

$$\text{Dep. Var.}_i = \beta_0 + \beta_1 \widehat{MR}_i + \Gamma' X_i + \epsilon_i. \quad (3)$$

The logic is that firms did not know in 2002 that the \$75 million threshold would determine 2004 compliance, so 2002 public float is less vulnerable to strategic manipulation.

4.3 Difference-in-differences and placebo checks

The paper also uses difference-in-differences comparisons around 2004, placebo cutoffs such as \$125 million and \$150 million, and pre-implementation tests in 2002 and 2003. These checks ask whether similar jumps existed before Section 404 compliance or at artificial thresholds.

4.4 Event studies

For market reactions, the paper studies SEC announcements and related news. The expected sign depends on the news content: if Section 404 is costly on net, then compliance delays or cost reductions should increase firm values, while strong enforcement news should reduce firm values. Abnormal returns are estimated using a market model with Fama-French and momentum factors.

4.5 Foreign-firm test

Foreign private issuers faced a separate cutoff. Firms above \$700 million in public float had to file auditor attestations earlier, while firms below the cutoff received a delay. This provides an additional test of whether auditor attestation itself raised costs and changed reporting.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies the local effect of Section 404 compliance for firms near the public-float cutoffs. The U.S. estimates are most directly about small domestic firms near \$75 million in public float, not necessarily the average large public company.

5.2 Why the design is credible

The threshold rule creates discontinuous compliance incentives. Firms just above and just below the cutoff are likely to be more comparable than firms far apart in size. The IV design strengthens this by using the 2002 cutoff, before firms could know exactly how later compliance would be tied to public float.

5.3 Key threat: manipulation of public float

The paper directly documents evidence of bunching below \$75 million in 2004. This means a simple OLS comparison may understate costs if firms expecting the largest burden deliberately avoided compliance. The IV estimates are therefore especially important.

5.4 Key threat: accelerated-filing deadline

Accelerated filer status also affected the timing of annual report filing. The paper addresses this by checking 2003, when accelerated filers faced earlier filing deadlines but not Section 404 compliance. The absence of similar effects in 2003 supports the interpretation that the 2004 jumps reflect Section 404 rather than accelerated filing alone.

5.5 How to interpret the net valuation result

The stock-return evidence does not mean that internal-control regulation has no benefits. Rather, it says that for the firms studied, the market appears to have believed the marginal cost of mandatory Section 404 compliance exceeded the marginal reporting-quality benefit.

6 Tables and figures

6.1 Table I: SEC Section 404 rules' effective dates

Read:

- Table I lays out the staggered implementation of accelerated filer status, management-report requirements, and annual-report deadlines.
- The crucial U.S. threshold is \$75 million in public float.
- Accelerated filers had to file management reports for fiscal years ending on or after November 15, 2004.

- Nonaccelerated firms received repeated delays, and auditor attestation for small firms was delayed even further.

Economic message: The table is the foundation of the identification strategy. The SEC rule created a regulatory discontinuity that can be used to compare treated and untreated firms near the cutoff.

6.2 Figure 1: Public-float manipulation around the cutoff

Read:

- Figure 1 plots the public-float distribution for firms that were not yet affected by the rule.
- There is little evidence of bunching below \$75 million in 2002 and 2003.
- In 2004, there is visible bunching just below \$75 million.
- This suggests that some firms may have adjusted public float to avoid Section 404 compliance once the rule became salient.

Economic message: The figure explains why the paper does not rely only on actual 2004 filing status. The manipulation evidence motivates the instrumental-variable strategy using the pre-terminated 2002 cutoff.

Table 1
SEC Section 404 Rules' Effective Dates

The SOX was signed into law on July 30, 2002. The SEC was put in charge of implementing Section 404. All U.S. incorporated companies had to finally submit a management report (MR) for fiscal year 2007, but non-accelerated firms were exempt from the auditor's attestation of the MR until June 2010. See the Internet Appendix for the SEC releases.

Rules Effective for Fiscal Years Ending On or After	Accelerated Filer Status	Management's Report Under Section 404	Annual Report Filing Deadline
Prior to December 2002	No filer status mandated by SEC.	Not required.	The annual report deadline is 90 days after the fiscal-year end. The reporting deadline stays at 90 days.
December 15, 2002	Companies with a public float of more than \$75 million become accelerated filers. (SEC Rel. 33-8128, September 5, 2002)	Not required.	
December 15, 2003	Companies with a public float of more than \$75 million become accelerated filers. Companies that were accelerated filers in the previous year keep their accelerated filer status.	Not required.	The annual report deadline for accelerated filers is changed to 75 days after the fiscal-year end. Non-accelerated filers' deadline remains at 90 days. (SEC Rel. 33-8128, September 5, 2002 and SEC Rel. 33-8307, Nov. 17, 2004)
November 15, 2004	Companies with a public float of more than \$75 million become accelerated filers. Companies that were accelerated filers in the previous year keep their accelerated filer status.	All companies that are accelerated filers have to file a management's report. (SEC Rel. 33-8286, Jan. 5, 2005 and Rel. 33-8392, February 24, 2004)	Non-accelerated filers: 90 days/Accelerated filers: 75 days.
December 15, 2005	Companies with a public float of more than \$75 million become accelerated filers. Companies that were accelerated filers in the previous year remain accelerated filers if their public float exceeds \$50 million, or change status to non-accelerated filers if their public float is less than \$50 million. (SEC Rel. 33-8644, December 21, 2005)	All companies that are accelerated filers have to file a management's report. Non-accelerated filers do not have to file a management's report.	

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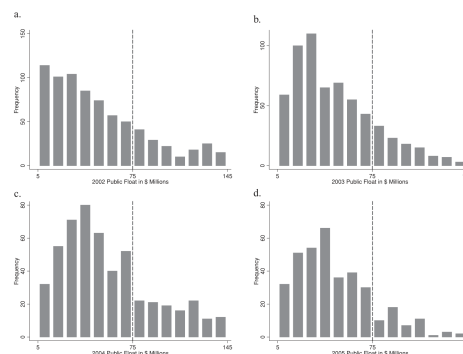


Figure 1. Did companies manipulate their public float to evade compliance? This figure depicts the public float distribution of firms not affected by the rule up to the reported year. These firms had a public float of less than \$75 million up to the year plotted in each graph. These firms being above the \$75 million cutoff means that they switch status to accelerated filer. Accelerated filers in years 2004 and 2005 had

Table I: SEC Section 404 rules' effective dates.

Figure 1: Public-float distribution around the \$75 million cutoff.

6.3 Figure 2 and Table II: Audit-fee costs

Read Figure 2:

- Audit fees rise sharply for accelerated filers in 2004, the first compliance year.
- The jump is visible both in the full sample and in the constant sample.
- The gap between accelerated and nonaccelerated filers remains high after 2004.

Read Table II:

- The dependent variable is the log of total audit fees in 2004.

- OLS estimates imply an audit-fee increase of roughly \$528 thousand to \$614 thousand for the mean firm.
- IV estimates are larger, implying an increase of about \$698 thousand to \$831 thousand.
- The first-stage relationship between the 2002 cutoff and 2004 compliance is strong.

Economic message: Section 404 imposed large direct compliance costs on small firms. The larger IV estimates are consistent with the idea that firms expecting the highest costs were especially likely to avoid crossing the cutoff.

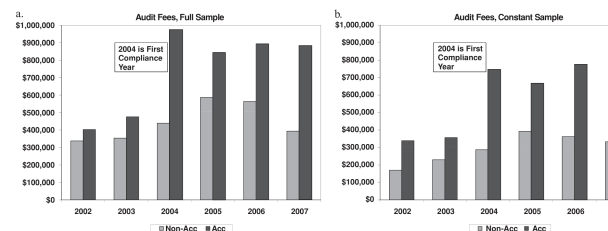


Figure 2. Audit fees annual means. The *Full Sample* graph uses all companies with a public float between \$50 million and \$100 million over the 2002 to 2007 period, with separate mean audit fees for accelerated (Acc) filers ($N = 174$ in 2004) and non-accelerated (Non-Acc) filers (171 in 2004). The *Constant Sample* graph further restricts attention to companies that have data for all 6 years and that filed a management (MR) for all 4 years of compliance ($N = 69$) or did not file an MR for all 4 years of compliance ($N = 21$). Starting in 2004, accelerated filers comply with Section 404. *Audit Fees* are mean audit and audit-related fees paid by the company to all its auditors for the relevant fiscal year.

Figure 2: Audit fees before and after the first compliance year.

Table II
Audit Fees Regressions for Fiscal Year 2004

This table presents the audit fees results. All OLS regressions are estimated with the model defined in Section II, equation (1); the IV regressions are estimated with the IV model defined in Section II, equation (3). The sample in estimations (1)–(4) covers all companies that had a public float between \$50 and \$100 million in 2004. The dependent variable is the logarithm of *Audit Fees*. In estimations (1) and (2) *MR* is a dummy variable equal to one if the company filed a management report (MR) in 2004, and in estimations (3) and (4) *MR* is the predicted treatment based on the first-stage regression; the IV estimation uses the instrument *PFL75*₂₀₀₂, a dummy equal to one if the company public float was above \$75 million in 2002. See the Appendix at the end of the main article for variables' definitions; *Market Size of Equity* is measured in billions. The regressions include (but are not reported here) a constant term; the OLS estimations include linear, quadratic, and cubic terms of *Public Float*, the public float reported in the annual reports. Regressions (2) and (4) control for industry fixed effects based on the Fama-French 12 sector definitions. The first-stage regressions have the same controls and fixed effects as the second stage. *First Stage Partial F-test* reports the *F*-test of the instrument in the first-stage regression. *Implied Audit Fee Increase* refers to the implied audit fee increase as a result of filing an MR for the mean company in \$ thousands. Robust *t*-statistics are reported in brackets. *, **, and *** denote two-sided statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent Variable Estimation Type	Log Total Audit Fees 2004			
	(1) OLS	(2) OLS	(3) IV	(4) IV
MR in fiscal year 2004	0.866*** [7.57]	0.744*** [7.39]	1.171*** [4.95]	0.983*** [3.65]
Log sales 2004		0.031 [1.09]		0.034 [1.09]
Log assets 2004		0.235*** [3.35]		0.218*** [2.79]
Log market size of equity 2003		0.050 [0.51]		-0.052 [-0.38]
Leverage 2004		0.612*** [2.62]		0.647*** [2.75]
Receivables scaled by total assets 2004		0.086 [0.35]		0.129 [0.55]
Big auditor 2004		0.370*** [3.94]		0.373*** [3.86]
Number of business segments 2004		0.040 [1.45]		0.047* [1.71]
Number of geographic segments 2004		0.070*** [2.91]		0.069*** [2.77]
Public float terms	Yes	Yes	No	No
Industry fixed effects	Yes	Yes	Yes	Yes
Observations	281	281	281	281
R^2	0.32	0.55	0.28	0.54
Implied audit fee increase	614.24	528.11	830.78	697.89
First-Stage Regression, <i>MR</i> instrumented by <i>PFL</i> ₂₀₀₂ > \$75 million, including first-stage controls, public float terms, and fixed effects				
<i>PFL</i> ₂₀₀₂ > \$75 Million			0.466*** [10.15]	0.376*** [7.74]
First-stage R^2			0.21	0.33
First-stage partial <i>F</i> -test			103.10	59.91

Table II: Audit-fee regressions for fiscal year 2004.

6.4 Figure 3 and Table III: Earnings management and accruals

Read Figure 3:

- The figure compares EPS and changes in EPS for firms that filed and did not file a management report.
- MR filers are more likely to report negative EPS.

- The distribution is consistent with reduced ability or willingness to manage earnings around common benchmarks.

Read Table III:

- Table III reports the effect of MR filing on accruals and discretionary accruals.
- MR filing lowers both accruals and discretionary accruals.
- The preferred IV estimates are larger than OLS: discretionary accruals fall by about 10.8 percentage points in the most complete IV specification.
- The implied earnings impact is economically large, reaching about \$15.1 million lower discretionary accruals for the mean firm.

Economic message: Section 404 improved measured earnings quality by making reporting more conservative. This is the main benefit side of the cost-benefit tradeoff.

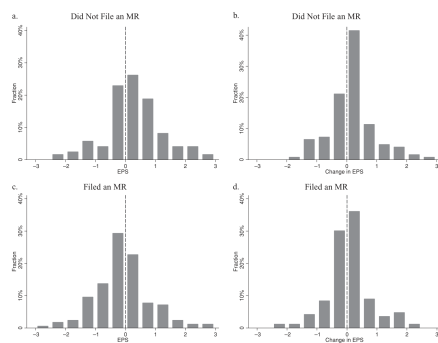


Figure 3: EPS and Change in EPS Distributions in 2004. The left graphs show earnings per share (EPS excluding extraordinary items) and the right graphs show the difference between the EPS in 2004 and the EPS in 2003 in dollars. The top graphs show the distribution of companies that did not file a management report (MR) and the bottom graphs show the distribution of companies that filed MR. All graphs use company data from a public float between \$50 million and \$100 million. The graphs are based on 166 MR filers and 123 non-filers.

Figure 3: EPS and change-in-EPS distributions in 2004.

Table III
Accruals and Discretionary Accruals Regressions in Fiscal Year 2004

This table presents the discretionary accruals results. All OLS regressions are estimated with the model defined in Section II, equation (1); the IV regressions are estimated with the IV model defined in Section II, equation (5). The sample consists of all companies that had a public float between \$50 and \$100 million in 2004. The dependent variables are *Accruals* and *Discretionary Accruals*, defined in the Internet Appendix. In estimations (1)/(2), (5), and (6) *MR* is a dummy variable equal to one if the company filed a management report in 2004; in the IV estimations (3)/(4), (7), and (8) *MR* is the predicted treatment based on the first-stage regression; see the Appendix at the end of the main article for variable definitions. *Market Size of Equity* is in billions. The regressions include (but are not reported here) a constant term and linear, quadratic, and cubic terms of *Public Float*. Industry fixed effects are based on the Fama-French 12 sector definitions. The IV estimation uses the instrument *PPL75₂₀₀₂*, a dummy equal to one if the company public float was above \$75 million in 2002. The first stage regressions have the same controls and fixed effects as the second stage. *First Stage Partial F-test* reports the *F*-test of the instrument in the first-stage regression. Robust *t*-statistics are reported in brackets. *, **, and *** denote two-sided statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent Variable	Accruals				Discretionary Accruals			
	(1) OLS	(2) OLS	(3) IV	(4) IV	(5) OLS	(6) OLS	(7) IV	(8) IV
MR in fiscal year 2004	-0.042** [-2.50]	-0.035* [-1.90]	-0.060* [-1.68]	-0.092** [-2.23]	-0.0475*** [-3.092]	-0.039** [-2.23]	-0.074** [-2.25]	-0.108*** [-2.70]
Cash flow from operations 2004	-0.052 [-0.83]	-0.052 [-0.83]	-0.049 [-0.69]	-0.049 [-0.69]	-0.069 [-1.06]	-0.069 [-1.06]	-0.068 [-0.88]	-0.068 [-0.88]
Change in net income 2004	0.213*** [3.00]	0.213*** [3.00]	0.207*** [2.82]	0.207*** [2.82]	0.219*** [3.22]	0.219*** [3.22]	0.211*** [3.22]	0.211*** [3.22]
Negative cash flow 2004	0.009 [0.41]	0.009 [0.41]	0.020 [0.89]	0.020 [0.89]	-0.016 [-0.66]	-0.016 [-0.66]	-0.005 [-0.20]	-0.005 [-0.20]
Book to market 2004	0.043** [2.02]	0.043** [2.02]	0.040** [2.07]	0.040** [2.07]	0.044** [2.23]	0.044** [2.23]	0.043** [2.34]	0.043** [2.34]
Big auditor 2004	-0.006 [-0.10]	-0.006 [-0.10]	-0.004 [-0.20]	-0.004 [-0.20]	0.015 [0.77]	0.015 [0.77]	0.020 [0.80]	0.020 [0.80]
Market size of equity 2003	-0.189 [-1.69]	-0.189 [-1.69]	0.065 [0.41]	0.065 [0.41]	-0.202 [-1.60]	-0.202 [-1.60]	0.086 [0.53]	0.086 [0.53]

Table III, part 1: Accruals and discretionary-accruals regressions.

6.5 Table III continued and Table IV: Event-study evidence

Read Table III continued:

- The continuation reports model fit, implied magnitudes, and first-stage statistics.
- The first-stage partial *F*-tests are large, supporting the relevance of the 2002 cutoff instrument.

Read Table IV:

- Table IV studies 10 SEC and news events related to Section 404 implementation.
- News that delayed or softened compliance generally produced positive abnormal returns for firms exposed to compliance.
- News that reaffirmed enforcement or gave relief mainly to nonfilers produced negative relative returns for MR filers.
- The signs are broadly consistent with the market viewing compliance as costly on net.

Economic message: The event studies translate the audit-fee and earnings-quality results into market-value terms. Investors appear to have priced Section 404 as a net burden for small firms.

Table III—Continued

Dependent Variable Estimation Type	Accruals				Discretionary Accruals			
	(1) OLS	(2) OLS	(3) IV	(4) IV	(5) OLS	(6) OLS	(7) IV	(8) IV
Public float terms	Yes	Yes	No	No	Yes	Yes	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	No	No	No	No
Observations	251	251	251	251	251	251	251	251
R ²	0.12	0.34	0.11	0.27	0.021	0.27	0.00	0.17
MR magnitude	−5.85	−4.98	−8.44	−12.93	−6.693	−5.46	−10.44	−15.14
First-Stage Regression, MR instrumented by PPL ₂₀₀₂ > \$75 million, including first-stage controls, public float terms, and fixed effects								
PPL ₂₀₀₂ > \$75 Million			0.442*** (9.17)	0.386*** (7.65)			0.456*** (10.06)	0.394*** (8.08)
First-stage R ²			0.19	0.29			0.17	0.28
First-stage partial F-test			84.17	58.59			101.28	65.23

Table III, part 2: First-stage and magnitude estimates.

**Table IV
Event Study Estimations**

This table presents the event study results. The Hypothesis columns report the expected market responses based on the hypothesis that delays in Section 404 compliance increase market value whereas news of SEC determination to enforce the rule decrease market valuation (+ denotes expected positive reaction, − expected negative reaction, none denotes no expected effect, and ± denotes unclear expected effect). Management report (MR) filers refers to the equal-weighted portfolio that buys all companies that filed an MR and were in a \$50 to \$100 million band in 2004. Nonfilers refers to the equal-weighted portfolio that buys all companies that did not file an MR and were in the same band. MR minus no-MR refers to the long-short portfolio. The estimations use a 120-day estimation window immediately before the event window. I estimate the market model: $R_{it} = \alpha_i + \beta_1 MKTRP_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \epsilon_{it}$, $E(\epsilon_{it}) = 0$, $\text{var}(\epsilon_{it}) = \sigma^2$ for the 120-day estimation window immediately before the 3-day event window, where R_{it} is the portfolio return and $MKTRP_t$, SMB_t , HML_t , and UMD_t are the returns on the market, the Fama-French size, book-to-market, and momentum factors. I use the predicted normal portfolio returns for the event window to calculate cumulative abnormal returns. For large estimation windows, the cumulative abnormal returns are distributed normally with expected value of zero and variance of $3 \cdot \sigma^2$ (MacKinlay (1997)). Two-sided t -statistics are reported in brackets. +, −, and ± denote two-sided statistical significance at the 10%, 5%, and 1% levels, respectively.

#	Event Date	Event Description [Expected Effect]	MR filers		Non-filers		MR minus no-MR	
			Hypothesis	Actual	Hypothesis	Actual	Hypothesis	Actual
(1)	August 27, 2002	SEC Press Release No. 2002-128: Accelerated filer status definition and acceleration schedule. [no effect expected because MR filing is not yet tied to acceleration rule]	none	−0.42% [−0.62]	none	−0.67% [−0.62]	none	0.25% [0.23]
(2)	May 28, 2003	SEC Press Release No. 2003-46 & Wall Street Journal article: “SEC Beta’s New Rule Aimed at Companies’ Internal Controls.” [setting the deadline, market expected earlier deadline]	+	2.96% [3.57]***	+	1.90% [2.23]**	±	1.05% [1.21]
(3)	May 6, 2003	SEC Final Rule No. 33-8268: Rule giving 1-year delay of Section 404 compliance for accelerated filers and 2 years for non-accelerated filers. [setting the deadline, market expected earlier deadline]	+	3.36% [3.86]***	+	2.44% [2.83]**	±	0.92% [1.07]
(4)	February 24, 2004	SEC Press Release No. 2004-21: 3-month extension of the compliance dates, requirement for companies to include an MR and an auditor’s attestation in the annual report. [same relief, but re-states the December 2004 deadline for most MR firms]	−	−1.57% [−2.17]**	+	−0.87% [−1.03]	−	−0.70% [−0.88]

Table IV, part 1: Event-study estimates.

6.6 Table IV continued and Table V: Longer-run stock returns

Read Table IV continued:

- Later 2004 and 2005 events include deadline extensions, advisory committee news, and announcements about reducing auditing costs.
- The MR-filer portfolio generally benefits from immediate relief, while nonfilers benefit from extensions that delay their future compliance.

Read Table V:

- Table V estimates a Fama-French-plus-momentum model for a long-short portfolio from May 2003 to June 2005.
- The portfolio longs predicted compliers and shorts predicted noncompliers.
- The monthly abnormal return is −0.81%, statistically significant.
- This implies about −17.7% abnormal buy-and-hold underperformance for predicted compliers over the implementation period.

Economic message: The longer-window return evidence reinforces the event-study evidence: the market valued Section 404 compliance as costly for small firms.

Table IV—Continued

#	Event Date	Event Description (Expected Effect)	MR filers		Non-filers		MR minus non-MR	
			Hypothesis	Actual	Hypothesis	Actual	Hypothesis	Actual
(5)	August 26, 2004	SEC Press Release No. 2004-121: SEC proposes accelerated filer status deadline change giving extra time for MR filers to comply. [non-filer SEC stance, immediate relief for MR filers]	+	1.44% (1.74)**	+	0.02%+ (-1.02)	+	1.46% (1.89)
(6)	November 17, 2004	SEC Press Release No. 2004-108: SEC announces relaxed accelerated filer deadline (remaining at 75 days instead of changing to 60). [non-filer SEC stance, immediate relief for MR filers]	+	1.97% (2.58)**	+	0.81%+ (1.26)	+	1.16% (1.60)
(7)	December 1, 2004	SEC Press Release No. 2004-162 & Wall Street Journal article: Accelerated filers with public float of over \$700 million get a 45-day postponement of filing an MR. [non-filer SEC stance, immediate relief for MR filers]	+	1.99% (2.57)**	+	0.85%+ (1.33)	+	1.14% (1.60)
(8)	December 16, 2004	SEC Press Release No. 2004-174: SEC establishes advisory committee to examine impact of Sarbanes-Oxley Act on smaller public companies. (SEC can reduce costs, direct benefit for MR filers)	+	2.04% (2.53)**	+	0.72%+ (1.10)	+	1.32% (1.87)
(9)	March 2, 2005	SEC Press Release No. 2005-25: Compliance dates for non-accelerated filers are extended to first annual ending on or after July 15, 2006. [non-filer get a filing extension]	±	-0.18% (-0.22)	+	1.61% (1.61)	+	-1.78% (-1.72)**
(10)	May 17, 2005	SEC Press Release No. 2005-74 and New York Times article: Regulators seek to trim cost of rules on	+	-0.83% (-1.12)	+	0.82%+ (1.06)	+	-1.76% (-2.51)**

The Effect of SOX Section 404

Table V
Two-Year Returns

This table presents monthly Fama-French estimations for the period from May 2003 to June 2005. $R_{p,t}$ is the monthly return of a long-short equal-weighted portfolio. The portfolio longs the industry-adjusted returns of companies that were predicted to comply with Section 404 based on their 2002 public float and their industry returns. The portfolio shorts the industry-adjusted returns of companies predicted not to comply with Section 404 based on their 2002 public float and their industry returns. All firms had to have 36 months of return data in CRSP to predict their public float in 2003 and 2004. Industry-adjusted firm returns in excess of $\pm 50\%$ are not included in the portfolio. $R_{f,t}$ is the risk-free return. $(R_{m,t} - R_{f,t})$ is the excess return on the market, calculated as the equal-weighted return on all NYSE, Amex, and NASDAQ stocks (from CRSP) minus the 1-month Treasury bill rate. *HML* (High Minus Low) is the average return on the two value portfolios minus the average return on the two growth portfolios. *Small Minus Big* (SMB) is the average return on the three small portfolios minus the average return on the three big portfolios. *MOM* is the momentum factor, the average return on the two high prior return portfolios minus the average return on the two low prior return portfolios. *HML*, *SMB*, and *MOM* come from Kenneth French's web site at Dartmouth via WRDS. *t*-statistics are reported in brackets. **, and *** denote two-sided statistical significance at the 10%, 5%, and 1% levels, respectively.

$R_{p,t} - R_{f,t} =$	-	0.0081 [-2.21]**	+	0.110 · ($R_{m,t} - R_{f,t}$) [0.98]	-	0.0346 · SMB_t [-0.25]
	+	0.331 · HML_t [1.88]*	+	0.030 · MOM_t [0.32]		($R^2 = 21.8\%$)

Table IV, part 2: Continued event-study estimates.

Table V: Two-year return evidence.

6.7 Foreign firms and robustness

Read:

- Foreign private issuers provide a second setting because the SEC delayed auditor attestation for foreign firms below \$700 million in public float.
- Foreign firms subject to auditor attestation paid higher audit fees and reported lower accruals and discretionary accruals.
- Event evidence for foreign firms is consistent with the U.S. evidence: delays are good news and enforcement is bad news for affected firms.

Economic message: The foreign-firm evidence suggests that the auditor-attestation component of Section 404 was an important source of costs, and that the main U.S. results are not merely an artifact of the domestic cutoff.

6.8 Appendix: Variable definitions

Read:

- The appendix defines the main variables used in the empirical tests, including public float, audit fees, earnings, leverage, accruals, big auditor status, and business/geographic segments.
- The definitions clarify the scaling of accounting variables and the construction of key controls.

Economic message: The appendix is useful because the paper’s estimates depend on careful measurement of public float, audit fees, and accruals. These definitions make the empirical design replicable.

Appendix

Variable Definitions

Variable	Units	Description (Compustat data items are in parentheses)
Public float	\$ millions	Public float reported on the company's annual report.
Market size of equity	\$ millions	Fiscal year-end stock price multiplied by the number of shares outstanding ($PRCC.F \times CSHO$).
Assets	\$ millions	Total Assets (AT).
Sales	\$ millions	Net Sales ($SALE$).
BE/ME	Ratio	Total Common Equity (CEQ) plus Deferred Taxes & Invest Tax Credit ($TXDITC$) (if available) minus Preferred Stock – Redemption ($PSTKRV$), Liquidating ($PSTKL$) or Carrying Value ($UPSTK$), used in that order, divided by the market value of equity at the end of the fiscal year ($PRCC.F \times CSHO$).
Audit fees	\$ millions	Total audit and audit-related fees from Audit Analytics.
Earnings	Ratio	IBC scaled by previous-year assets (AT). Values in excess of $\pm 100\%$ were censored.
Leverage	Ratio	Long-term debt ($DLTT$) + Debt in Current Liabilities (DLC) divided by Long-term debt ($DLTT$) + Debt in Current liabilities (DLC) + Market Size of Equity.
IPO year	Year	Company Initial Public Offering Date ($IPODATE$).
Receivables	Ratio	Inventories ($INVT$) plus Receivables ($RECT$) scaled by Total Assets (AT).
Cash flow from operations	Ratio	Operating Cash Flow from continuing operations ($XIDOC - OANCF$) scaled by previous-year assets (TA). Values in excess of $\pm 100\%$ were censored.
Change in net income	Ratio	Change in Net Income (NI) scaled by previous-year assets (TA).
Negative cash flow in 2003	Y/N	Dummy variable equal to one if previous-year Cash Flow from Operations is negative.
Accruals	Ratio	Total Accruals are IBC minus cash flow from continuing operations ($IBC - OANCF + XIDOC$) scaled by previous-year assets (AT). Values in excess of $\pm 100\%$ were censored.
Big auditor	Y/N	Dummy variable equal to one if the company used one of the big auditing firms (Deloitte and Touche, Ernst and Young, KPMG, PricewaterhouseCooper).
Number of business segments	Integer	Number of business segments from Compustat Segments.
Number of geographic segments	Integer	Number of geographic segments from Compustat Segments.
NASDAQ listed	Y/N	Dummy variable equal to one if the company is traded on NASDAQ.

Appendix: Variable definitions.

7 Short summary

- The paper studies the effect of SOX Section 404 on small public firms.
- The key empirical strategy is a regression discontinuity around SEC public-float cutoffs.
- U.S. firms above \$75 million in public float had to comply earlier; firms below could delay.
- The paper documents bunching below the cutoff in 2004, suggesting possible avoidance behavior.
- To address this, the paper instruments actual 2004 filing with whether the firm exceeded the cutoff in 2002.
- Section 404 substantially increased audit fees.

- Section 404 reduced accruals and discretionary accruals, consistent with more conservative reporting.
- Event studies show positive market reactions to compliance delays and negative reactions to strict enforcement news.
- A long-short portfolio of predicted compliers versus noncompliers earns a significantly negative abnormal return during implementation.
- Foreign-firm evidence around a \$700 million cutoff supports the interpretation that auditor attestation was costly.
- The main conclusion is that Section 404 improved reporting quality but reduced the market value of small firms on net.

8 Conclusion

8.1 Core conclusion

Iliev shows that SOX Section 404 had both benefits and costs. Compliance increased audit fees and other likely compliance costs, but it also reduced discretionary accruals and made reported earnings more conservative. For small firms near the compliance cutoff, the market-value evidence indicates that the costs outweighed the benefits.

8.2 Economic intuition

Internal-control regulation can improve financial reporting by forcing firms to document controls, test systems, and involve outside auditors. But these systems involve large fixed costs. For a small public firm with limited scale, a few hundred thousand dollars in extra audit fees plus internal compliance costs can be economically large. Thus, regulation that may be reasonable for large firms can be value-reducing for small firms.

8.3 Incremental contribution revisited

The paper's contribution is not simply to show that SOX was expensive. Its main contribution is to isolate the effect of Section 404 using a quasi-experimental design. The public-float cutoff allows the paper to separate Section 404 effects from other post-Enron changes in litigation risk, auditing norms, investor scrutiny, and corporate governance practices.

8.4 Implications

The findings support the idea that regulation should consider firm size and fixed costs. Uniform reporting rules may produce higher-quality disclosures, but the net welfare effect can differ sharply across firms. For small firms, exemptions, delayed compliance, or scaled requirements may be economically justified.

8.5 Limitations

The estimates are local to firms near the public-float thresholds. The paper does not imply that Section 404 reduced value for all firms, especially large firms. It also measures accounting outcomes and market reactions rather than directly observing all investor-protection benefits. Finally, the bunching evidence shows that regulatory thresholds can change firm behavior, which complicates interpretation of simple cutoff comparisons.

8.6 Takeaway

The enduring takeaway is that mandatory internal-control regulation can improve reported earnings quality while still lowering firm value when compliance costs are large. For small firms, SOX Section 404 appears to have been a costly form of financial reporting regulation whose benefits did not fully offset its costs.